

in most areas today, be it e-commerce, healthcare, customer service or automotives.

Machine learning (ML) tracks users' experiences and history through simple programming methods. Advancement of ML has increased the opportunities for specialisation in several new subsets, such as neural networks, natural language processing (NLP) and deep learning. Deep learning is a branch of ML where multi-layered neural network algorithms attempt to model high-level abstractions in data.

Information technology (IT) and cybersecurity firms have begun to adopt artificial neural networks to monitor and prevent distributed denial of services (DDoS) attacks. AI-powered platforms are set to help traffic management, too. By 2020, these two innovations will be deeply entrenched in both business and consumer activities.

Cloud computing

Cloud computing provides a gateway to AI tools to gear up the global technology market. Cloud computing's Pay As You Go model helps reduce the cost of investment in software and hardware. As part of digital transformation, more and more companies are expected to move their IT systems to the cloud by 2019. Everything as a service (XaaS) solutions, including software as a service (SaaS), infrastructure as a service (IaaS), function as a service (FaaS) and platform as a service (PaaS) will make the journey of technology faster to enhance customer experience.

The flexible consumption model of cloud computing will provide companies with the leverage of taking expertise from the world's biggest technology companies and startups. It will give them access to companies of all sizes and newer technologies, by including flexible consumption in their innovation strategies.

The new computing continuum is multi-cloud models with their diverse, decentralised and autonomic management, and hybrid cloud models



Cloud computing is changing the world of computers (Credit: www.sharda.ac.in)

that cross boundaries between internal and external cloud services, or between public, private and community providers.

In 2020, edge computing will be one of the the top technology trends enabling intelligent networks where connected devices will perform necessary analytics right at the location to perform specific actions. This will all happen within a few milliseconds with the help of hardware accelerators and highly-specialised computing devices.

The Internet of Things

The Internet of Things (IoT) is a system where devices are connected wirelessly with the help of unique identifiers. It enables transfer of data without human intervention. One of the best applications of the IoT is the safety measures to protect students. ID cards in some schools contain a chip to track the location of a child. It reduces the chances of kidnapping or a child going missing.

Farmers can now solve problems with the help of cameras, sensors and weather stations. Apps are being developed to help farmers check the health of their crops and know how much to irrigate and which crop is suitable for which land. This facilitates maximum yields.

IoT solutions are also influencing the hospitality industry.

The progress of the ambitious smart cities project is significant. The IoT has made it easier to collect data to

analyse and make better decisions based on quality and operational efficiency. Smart streetlights and fuel-monitoring systems are great examples of latest IoT applications.

In a smart home, sensors send collected data to smart home apps, check energy usage and manage it automatically. These detect issues, and inform concerned parties and even call for assistance, thus helping maintain home appliances easily.

Smart meters measure energy and water consumption, and send data over the connected app. Smart waste management helps keep cities clean. Users looking for parking are directed to the nearest available spot. Sensors or smart cards allow them to pay for parking automatically. Prepaid electricity for rural areas has paved the way for rural development.

Low power wide area network (LPWAN) wireless communication technology connects objects that require low-power, long-range and low data rate to the IoT. Many objects connected to the IoT only need to transfer small amounts of data, such as commands and status updates. This is where LPWAN becomes useful.

Be it a simple smartwatch or an ECG recorder, the medical industry has benefited majorly from technology. The IoT has made it easy to monitor health. It helps transmit signals directly to hospitals, making it easy to track in real